In [5]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
from textblob import TextBlob # to understand meaning of the sentences
```

In [6]:
```
#load the dataset
data=pd.read_csv("Sentiment_Analysis .csv")
data.head(10)
```

Out[6]:

	Sentences
0	The product's exceptional quality and performa...
1	Poor customer support and the product's subpar...
2	The service met standard requirements without ...
3	Outstanding customer service accompanied the p...
4	The product's failure to meet basic requiremen...
5	The service provided met basic needs without n...
6	Remarkable product quality and performance exc...
7	Poor customer service and a faulty product led...
8	The service provided was average, meeting mini...
9	The purchase proved delightful, with the produ...

In [7]:
```
# define the sentiment analysis function
def get_sentiment(text):
    meaning=TextBlob(text)

    if meaning.sentiment.polarity>0:
        return "Positive"
    elif meaning.sentiment.polarity<0:
        return "Negative"
    else:
```

```
        return "Neutral"
```

In [8]:
```
data["Sentiment"]=data["Sentences"].apply(get_sentiment)
```

In [9]:
```
%pip install openpyxl
data.to_excel("SentimentAnalysisbyYatin.xlsx",index=False)
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: openpyxl in C:\Users\Y.K\AppData\Roaming\Python\Python314\site-packages (3.1.5)
Requirement already satisfied: et-xmlfile in C:\Users\Y.K\AppData\Roaming\Python\Python314\site-packages (from openpy
xl) (2.0.0)
Note: you may need to restart the kernel to use updated packages.
```

In [10]:
```
data.head()
```

Out[10]:

	Sentences	Sentiment
0	The product's exceptional quality and performa...	Positive
1	Poor customer support and the product's subpar...	Negative
2	The service met standard requirements without ...	Positive
3	Outstanding customer service accompanied the p...	Positive
4	The product's failure to meet basic requiremen...	Negative

In [11]:
```
#filter only negative views
negative_reviews=data[data["Sentiment"]=="Negative"]
negative_reviews.head(10)
```

Out[11]:

	Sentences	Sentiment
1	Poor customer support and the product's subpar...	Negative
4	The product's failure to meet basic requiremen...	Negative
7	Poor customer service and a faulty product led...	Negative
8	The service provided was average, meeting mini...	Negative
13	Dreadful customer support coupled with a defec...	Negative
16	Severe product shortcomings led to considerabl...	Negative
17	The service was standard, meeting basic needs ...	Negative
19	Disappointing product quality and poor custome...	Negative
20	The service provided was average, meeting basi...	Negative

In [12]:
```
#combine all the negative sentences into one string/paragraph
negative_para=" ".join(negative_reviews["Sentences"])
print(negative_para)
```

```
Poor customer support and the product's subpar performance led to frustration and disappointment, regretting the purc
hase entirely. The product's failure to meet basic requirements resulted in immense inconvenience and frustration, ca
using regret over the purchase. Poor customer service and a faulty product led to significant inconvenience and frust
ration, resulting in regret over the purchase decision. The service provided was average, meeting minimum requirement
s without standout features, resulting in a standard user experience. Dreadful customer support coupled with a defect
ive product caused immense frustration and disappointment, regretting the purchase. Severe product shortcomings led t
o considerable dissatisfaction and regret over the purchase due to a lack of promised functionality. The service was
standard, meeting basic needs without noteworthy features, resulting in an average user experience without extremes.
Disappointing product quality and poor customer service resulted in frustration and regret, tarnishing the overall pu
rchase experience. The service provided was average, meeting basic requirements without standout features, resulting
in an unremarkable user experience.
```

In [13]:
```
# create a word cloud for negative paragraph
wordcloud=WordCloud(width=500,height=300,background_color="black").generate(negative_para)
plt.figure(figsize=(10,4))
plt.imshow(wordcloud,interpolation="bilinear")
plt.title("Most Common Words in NEGATIVE sentences")
plt.axis("off")
plt.show()
```

Most Common Words in NEGATIVE sentences

